Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_638216fd6a2940a38e670f79a095dc07
Date	2026-11-02 15:40:29
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
size_mismatch	major	99%	length 115.68 mm (nom 120.00 ±0.10)
size_mismatch	major	99%	width 77.44 mm (nom 80.00 ±0.10)
size_mismatch	major	99%	hole dia 10.86 (nom 12.00 ±0.05), position off 1.24 mm (±0.10)
size_mismatch	major	99%	hole dia 11.21 (nom 12.00 ±0.05), position off 3.05 mm (±0.10)
size_mismatch	major	99%	hole dia 10.49 (nom 12.00 ±0.05), position off 2.87 mm (±0.10)
size_mismatch	major	99%	hole dia 10.75 (nom 12.00 ±0.05), position off 5.49 mm (±0.10)
dent	major	95%	~18 mm

2. Root cause

Cause: Improper Bending Force or Over Bending

Reason: The significant deviation in part length (115.68mm vs 120.00mm) and width (77.44mm vs 80.00mm) suggests incorrect bending parameters or excessive force application during the forming process [seed-001].

Effect: Dimensional Inaccuracy

Confidence: 70%

Supporting evidence
- **seed-001**: Improper Bending Force or Over Bending causes deformation and dimensional inaccuracy. Precautions: optimize bending parameters, use proper tooling, calibrate machines regularly.

3. Solution

Corrective actions (fix this occurrence)

1. Calibrate bending machine force settings
2. Adjust bending parameters to match material thickness

Preventive actions (stop it recurring)

1. Implement regular machine calibration schedule
2. Optimize bending parameters for SPCC steel

Precautions

- Monitor bending force during production runs

4. Other causes considered

Rank	Cause	Confidence	Reason

2	Tool Wear and Die Damage	67%	The presence of a ~18mm dent and significant hole diameter/positioning errors (10.49mm-11.21mm vs 12.00mm) indicates potential die degradation or tooling misalignment [seed-003].
3	Material Inconsistency and Hardness Variation	64%	The combination of hole position shifts and size mismatches can be exacerbated by variations in SPCC steel hardness, leading to unpredictable material flow during bending [seed-002].

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.